

## **Optimizing RAG Document Retrieval for Agronomic Advice**

### **Describe the problem that the selected project is trying to solve.**

Small-scale farmers and agricultural professionals need access to reliable information to address issues related to crop diseases, pests, nutritional deficiencies, soil management, fertilization, and climate change adaptation. Although this information exists in manuals, technical guides, and other specialized documents, quickly locating the most relevant content for a specific query can be challenging.

This difficulty delays decision-making and increases the risk of implementing inappropriate practices that affect agricultural productivity. Therefore, the project aims to improve the retrieval of relevant documents within a Retrieval-Augmented Generation (RAG) system, facilitating access to useful information that serves as a basis for providing well-founded answers.

### **Who is affected by this problem?**

- Small-scale farmers are the most affected, as they rely on timely information to protect their crops and maintain production.
- Rural extension workers who need quick tools to assist producers.
- Women farmers who, in many regions, have less access to formal education and are more affected by the information gap.
- Local ecosystems are indirectly affected by the misuse of inputs (fertilizers or pesticides applied incorrectly due to a lack of knowledge).

### **Why is solving this problem important?**

When farmers cannot quickly find relevant information, they may delay the identification of diseases or pests, apply inappropriate treatments, or use fertilizers and other agricultural practices incorrectly. These situations can lead to economic losses, reduce productivity, and affect crop sustainability. Facilitating access to pertinent information contributes to better decision-making and fosters more efficient and sustainable agricultural production.

### **Which values should guide your solution?**

- **Accessibility:** The tool must be useful for farmers with varying levels of digital literacy and across diverse socioeconomic contexts.

- **Reliability:** Prioritize up-to-date documents to prevent obsolete recommendations from reaching the farmer.
- **Transparency:** Display the sources of the information so that farmers or extension workers can verify it.
- **Reliability:** Prioritize information from reliable and recent sources using ranking and filtering mechanisms.
- **Transparency:** Provide traceability of responses by displaying the sources consulted to enhance transparency and user trust.

### **Why did you choose these values?**

These values ensure that the solution is not only technically efficient but also useful and responsible for those who rely on it. Accessing reliable information helps improve the quality of agricultural decisions, while transparency strengthens trust in the system by revealing the origin of the data used. Furthermore, accessibility and equity enable more people to benefit from the tool, regardless of their circumstances.

### **What ethical risks should your team consider?**

- Retrieving documents that are not truly relevant to the query.
- Basing answers on outdated information.
- Favoring certain types of documents over others without clear criteria.
- Potential biases present in the document database used.

### **How will your team reduce these risks?**

- Continuously evaluate retrieval performance using objective metrics
- Prioritize reliable and up-to-date sources by weighting document recency in the ranking process.
- Use clear, consistent ranking criteria (source reliability, recency, relevance) so results aren't skewed by hidden or arbitrary factors.
- Regularly review the document collection for coverage gaps (e.g., missing crops, regions, or languages) and prioritize diversifying sources where gaps are found.

### **What outcomes would define success?**

- Retrieve highly relevant documents for each query.
- Facilitate quick access to useful agricultural information.
- Contribute to better decision-making by farmers and extension workers.
- Build trust through well-founded and transparent information.

### **Final value-based problem statement**

Small-scale farmers and agricultural professionals often face difficulties accessing the most relevant agricultural information when making decisions about crop diseases, pests, soil management, fertilization, and climate change adaptation. Although valuable knowledge exists in technical manuals and extension documents, finding the right information quickly remains a challenge, increasing the risk of delayed or inappropriate decisions that can affect productivity and sustainability. Our project aims to improve document retrieval within a Retrieval-Augmented Generation (RAG) system, enabling users to identify the most relevant and reliable information for their needs. The solution is guided by the values of accessibility, transparency, reliability, and equity by prioritizing up-to-date sources and clearly presenting the supporting documents. By improving access to trustworthy agricultural knowledge, we aim to support informed decision-making and strengthen sustainable farming practices in rural communities.